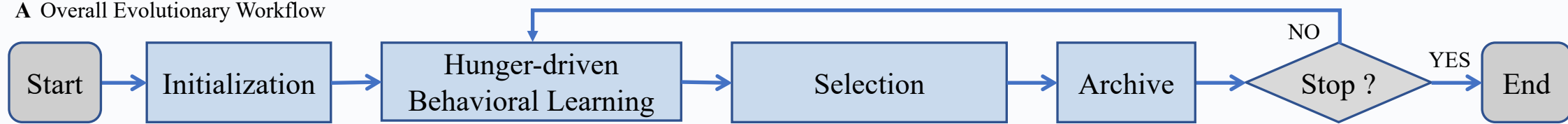
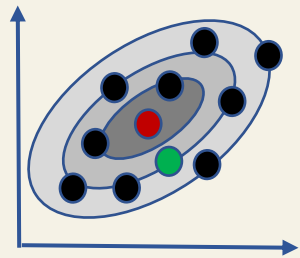


A Overall Evolutionary Workflow



B Hunger-driven Behavioral Learning

Random Initialization

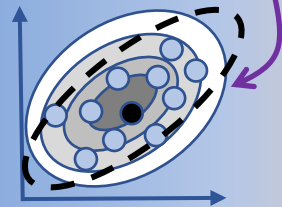


● x_{gbest} ● x_{pbest}
● x_i ● $x_i \cdot (1 + \text{randn})$



Noncooperative Mode

$$x = x_i \cdot (1 + \text{randn})$$



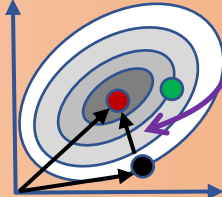
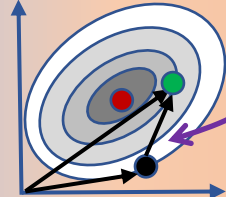
Exploration

YES

Cooperative Mode

$$x = \alpha \tilde{x}_r + R\beta |x_{pbest} - x_i|$$

$$x = \alpha \tilde{x}_r + R\beta |x_{gbest} - x_i|$$



Early Cooperative Phase ?

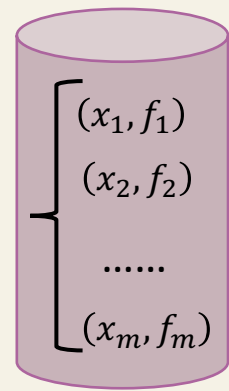
$r_{gate} \leq tvL ?$
 (Time-Varying Gate)

NO

Exploitation



Update Archive



$$A \leftarrow \text{ND}(A \cup P)$$